

TeachTrace AI

AI-Powered Classroom Intelligence & Learning Gap Detection System

Product Requirements Document (PRD)

Version: 1.0 **Project Type:** Hackathon MVP **Primary Users:** Teachers / Educators

Core Technologies: RAG, NLP/Linguistic Analysis, LLM, Python, Gradio/Streamlit, GitHub

1. Product Overview

TeachTrace AI is an AI-powered classroom intelligence system designed to help teachers understand **why students are struggling**, rather than simply identifying which students received low marks.

Teachers upload students' written answers, assignments, quizzes, or test papers. TeachTrace AI analyzes the responses using **Retrieval-Augmented Generation (RAG)** and linguistic analysis to identify recurring misconceptions, concept gaps, grammar and reasoning patterns, ambiguous questions, and class-wide learning trends.

The system transforms classroom evidence into actionable teaching decisions by generating a **Learning Gap Map**, recommending specific concepts to reteach, creating targeted intervention activities, generating follow-up assessments, and comparing results to determine whether student understanding improved.

2. Problem Statement

Traditional assessment systems mainly answer: **"Who got the lowest marks?"** They often fail to answer: **"Why are students getting the answers wrong?"**

Teachers may have to manually examine dozens of answers to discover recurring mistakes. This is time-consuming and makes it difficult to identify whether a problem comes from lack of conceptual understanding, repeated misconceptions, poor reasoning, language or grammar difficulties, misinterpretation of questions, ambiguous wording, or class-wide learning gaps.

3. Product Vision

To create an intelligent classroom analysis system that helps teachers move from **Assessment** → **Diagnosis** → **Intervention** → **Reassessment** instead of simply **Assessment** → **Marks**.

4. Target Users

Primary User — Teachers: Upload student responses, analyze classroom performance, identify common learning gaps, view trends, generate intervention activities, generate follow-up assessments, and compare before/after performance.

Secondary Users: Schools, tutors, academic coordinators, and online educators.

5. Goals

The MVP should: (1) allow teachers to upload answer data; (2) analyze answers using AI/NLP; (3) identify recurring misconceptions and concept gaps; (4) detect linguistic and reasoning patterns; (5) identify potentially ambiguous questions; (6) aggregate individual errors into class-wide trends; (7) generate a Learning Gap Map; (8) recommend what to reteach; (9) generate a targeted intervention; (10) generate a short follow-up assessment; (11) compare initial and follow-up results; and (12) present results through a simple teacher-friendly dashboard.

6. Core User Journey

Teacher → Uploads Test/Student Answers → TeachTrace AI → RAG + NLP Analysis → Individual Response Analysis → Classroom-Level Pattern Detection → Learning Gap Map → Reteaching Recommendations → Targeted Intervention Activity → Follow-up Assessment → Before vs After Comparison → Learning Improvement Report

7. Key Features

7.1 Student Response Upload

Teachers can upload CSV, TXT, or PDF data. For the MVP, text/CSV can be prioritized. Example fields: Student, Question, Answer.

7.2 RAG-Based Knowledge Retrieval

TeachTrace AI can use a knowledge base containing subject material, lesson objectives, textbook content, teacher-provided notes, correct concepts/answers, and learning objectives. RAG retrieves relevant information before the LLM analyzes student responses, helping the system compare answers against the actual teaching material.

8. Linguistic Analysis

The system analyzes written responses for **grammar patterns** (sentence structure, subject-verb agreement, tense errors, missing words), **vocabulary** (limited vocabulary, incorrect terminology, misuse of subject-specific terms), **reasoning** (unsupported conclusions, missing reasoning steps, contradictions, logical gaps), and **conceptual understanding** (correct, partial, misconception, or incorrect understanding).

9. Misconception Detection

The system identifies mistakes that occur repeatedly across students. Example: if multiple students say plants use sunlight to “breathe,” TeachTrace AI can identify the common misconception that students are confusing photosynthesis with respiration.

10. Learning Gap Map

The Learning Gap Map is the central output of TeachTrace AI. It shows the concept, number of students affected, severity, and detected issue.

Concept	Students Affected	Severity	Detected Issue
Photosynthesis	18/30	High	Major misconception
Respiration	12/30	Medium	Concept confusion
Scientific vocabulary	9/30	Medium	Terminology errors
Reasoning	7/30	Low	Incomplete explanations

11. Reteaching Recommendations

TeachTrace AI converts identified gaps into actionable recommendations. Example: if students confuse photosynthesis and respiration, the system recommends reteaching the difference using a short explanation, comparison diagram, real-world example, and misconception-based question.

12. Targeted Intervention Generator

The system automatically creates a short activity based on the detected learning gap. Example: a “Photosynthesis vs Respiration: Sort & Explain” activity where students classify statements and explain one choice.

13. Follow-Up Assessment Generator

After intervention, TeachTrace AI generates a short assessment focused on the previously detected learning gaps. This creates a direct connection between diagnosis, intervention, and reassessment.

14. Improvement Measurement

Teachers can upload follow-up results. TeachTrace AI compares before-intervention and after-intervention performance and can classify the outcome as significant improvement, moderate improvement, limited improvement, or no improvement.

15. Question Ambiguity Detection

TeachTrace AI can flag questions where an unusually high number of incorrect responses may be caused by the question itself. For example, if many students interpret the same question differently, the system can report **Potential Question Ambiguity** rather than automatically labeling the responses as learning failures.

16. Class-Wide Learning Trends

The system aggregates student-level findings into a classroom profile, such as conceptual gaps, reasoning issues, grammar issues, vocabulary issues, and question ambiguity. This gives teachers a quick overview of the class's learning needs.

17. Dashboard

The MVP dashboard should contain: **Upload** (student answers, reference material, subject/topic), **Analysis** (Analyze Responses), **Learning Gap Map** (top gaps, affected students, severity), **Recommendations** (what to reteach and suggested strategy), **Intervention** (generated classroom activity), **Follow-Up** (generated assessment and result upload), and **Improvement** (before/after performance and remaining gaps).

18. AI Output Example

After analyzing 30 students, the system might report: **Primary Learning Gap:** Students are confusing correlation with causation. **Affected:** 19/30. **Severity:** High. **Evidence:** recurring interpretations treating observational relationships as proof of cause. **Recommended Action:** reteach correlation vs causation using contrasting examples. **Intervention:** a short classification activity. **Follow-Up:** a 3-question conceptual assessment.

19. Technical Architecture

Flow: Teacher → Upload Interface (Gradio/Streamlit) → Document Parser → RAG Pipeline + Vector Store → LLM + Linguistic Analysis → Learning Gap Detection → Recommendations / Intervention / Assessment Generator → Teacher Dashboard.

20. Suggested Technology Stack

Frontend: Gradio or Streamlit

Backend: Python

LLM: Groq-hosted LLM

RAG: LangChain or LlamaIndex + Chroma/FAISS

NLP: Python NLP libraries + LLM-based linguistic analysis

Data Processing: Pandas + document parsing libraries

Version Control: GitHub

Deployment: Hugging Face Spaces or suitable cloud platform

21. MVP Scope

The hackathon MVP should focus on one strong workflow: **Upload classroom answers → Detect learning gaps → Generate intervention → Generate follow-up assessment → Measure improvement.**

MVP Input: Student answers + correct/reference material.

MVP Output: Individual response analysis, common misconceptions, Learning Gap Map, reteaching recommendations, targeted intervention, follow-up assessment, and improvement comparison.

22. Out of Scope for MVP

Student accounts; parent accounts; attendance management; complete school management system; automated grading for every subject; mobile application; real-time classroom monitoring; facial recognition; student surveillance; large-scale school analytics.

23. Success Metrics

Analysis Accuracy: How accurately the system identifies actual learning gaps.

Misconception Detection: Percentage of major recurring misconceptions correctly identified.

Actionability: Whether teachers find recommendations useful.

Intervention Relevance: Whether generated activities directly address detected gaps.

Learning Improvement: Difference between pre-intervention and post-intervention performance.

24. Example End-to-End Scenario

A teacher uploads 30 students' English comprehension answers. TeachTrace AI discovers that 17 students struggle with inference, 12 confuse main idea with supporting detail, 9 have recurring sentence-structure problems, and 6 misunderstand an ambiguous question.

The system identifies inference as the top priority, recommends reteaching inference from textual evidence, creates a short "Evidence or Guess?" activity, and generates three inference-based follow-up questions. If accuracy increases from 43% to 76%, the teacher can see that the intervention produced measurable improvement.

25. Unique Selling Proposition

Traditional Assessment: Test → Marks → Identify weak students

TeachTrace AI: Test → Analyze Responses → Discover Why → Map Learning Gaps → Reteach → Intervene → Reassess → Measure Improvement

The key differentiator is: **"TeachTrace AI doesn't just tell teachers who got the answer wrong. It tells them why the class got it wrong and what to do next."**

26. Hackathon Value Proposition

TeachTrace AI addresses a genuine education problem while combining RAG, LLM reasoning, NLP, linguistic analysis, pattern detection, data aggregation, AI-generated interventions, AI-generated assessments, and learning-outcome measurement. This makes it more than a simple chatbot or AI tutor.

27. Future Roadmap

Phase 1 — Hackathon MVP: Answer upload, RAG analysis, misconception detection, Learning Gap Map, intervention generation, follow-up assessment.

Phase 2: PDF/image handwritten-answer processing, subject-specific models, teacher accounts, historical class analytics.

Phase 3: School-wide analytics, personalized student learning profiles, curriculum alignment, multi-language support, and LMS integration.

28. Final Product Statement

TeachTrace AI is an AI-powered classroom intelligence system that transforms student answers into actionable teaching decisions. By combining RAG, linguistic analysis, and classroom-level pattern detection, it identifies not only what students got wrong but why they got it wrong, what teachers should reteach, which intervention to use, and whether learning actually improved.

Core Slogan

"Don't just trace the marks. Trace the learning gap."